Can Personality Predict Drug Use?
A Machine Learning Approach

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Introduction & Problem Statement

- Drug use is a growing global public health issue.
  - 292M users, +20% over 10 years
- Personality traits are known to be strong predictors of life outcomes.
- This project explores whether personality and demographic data can help predict drug use, improving targeted prevention and intervention strategies.

Research Objectives

- Predict drug use using personality and demographic characteristics.
- Examine whether predictive patterns differ across drug categories: stimulants, depressants, and hallucinogens.
- Compare performance of logistic regression and random forest models.
- Translate findings into public health recommendations.

Dataset Overview

Source: UCI Machine Learning Repository

Respondents: 1,885 individuals.

Variables: Demographics (age, gender, education, country, ethnicity), Big Five traits, impulsivity, sensation seeking, and self-reported drug use (18 substances).

7 Class Classfication for Each Drug	
Value	**Class**
CL0	Never Used
CL1	Used over a Decade Ago
CL2	Used in Last Decade
CL3	Used in Last Year
CL4	Used in Last Month
CL5	Used in Last Week
CL6	Used in Last Day

Data Preparation

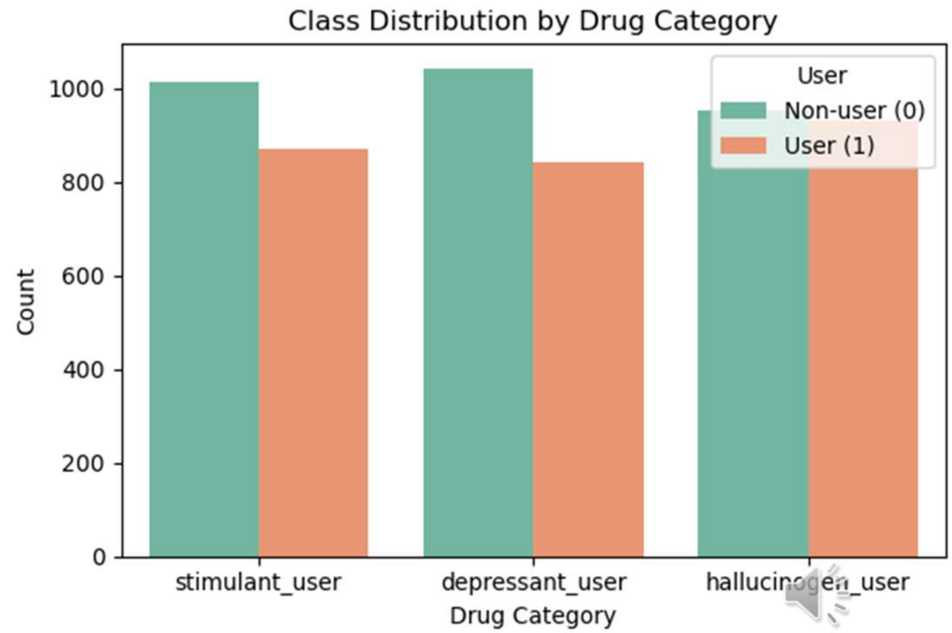
Converted drug usage into binary (user vs. non-user).

Grouped Targets:

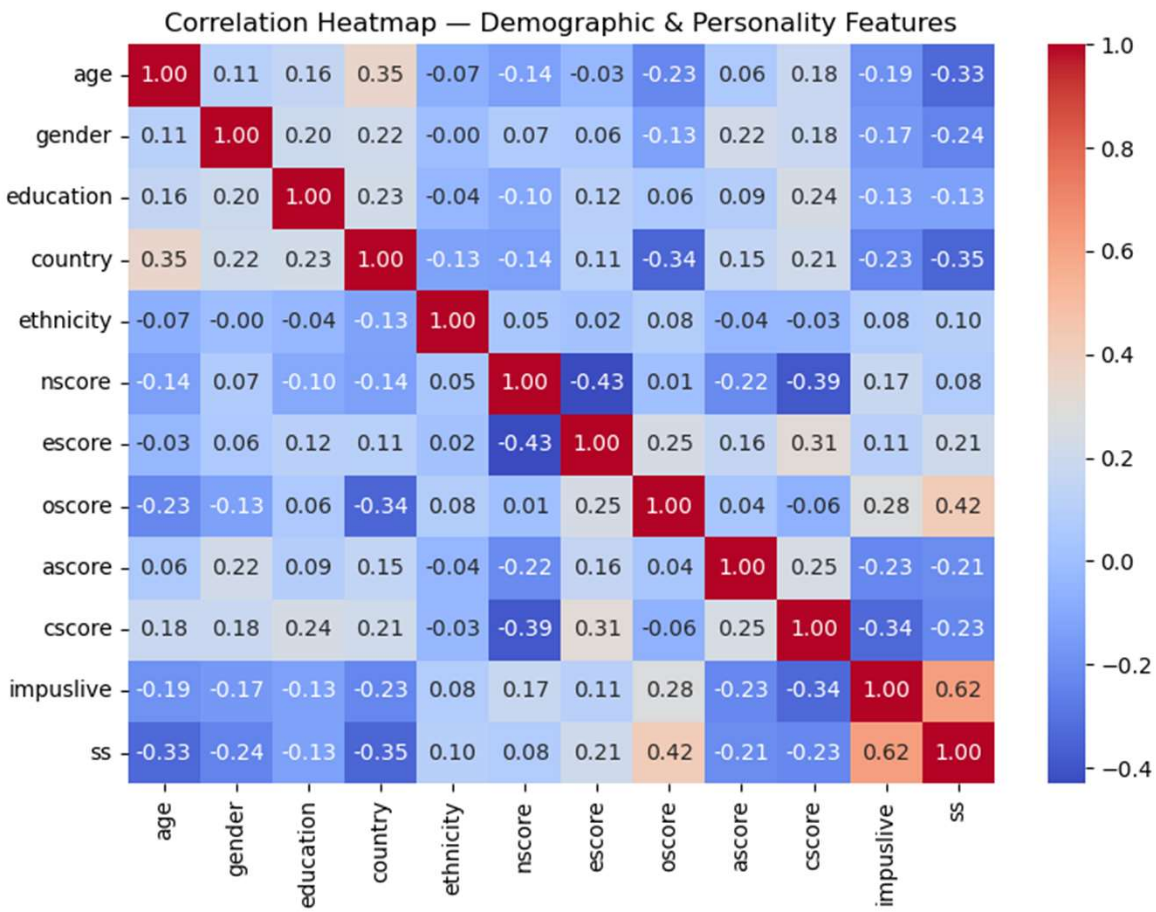
- Stimulants (e.g., amphetamines, cocaine)
- Depressants (e.g., benzodiazepines, heroin)
- Hallucinogens (e.g., LSD, ecstasy, mushrooms)

Scaled numeric features using StandardScaler.

Addressed class imbalance using SMOTE for balanced training data.



Data Exploration



Correlation heatmap of numerical features showing moderate correlations between some traits but no multicollinearity (|r| < 0.8)

Modeling Approach

- Logistic Regression: Baseline model for interpretable coefficients.
- Random Forest: Captures nonlinear relationships and provides feature importance rankings.
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, PR AUC.

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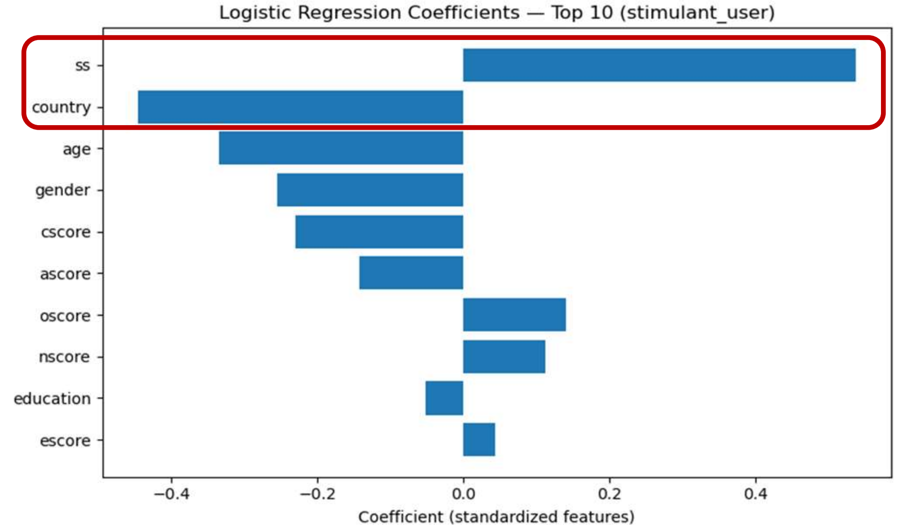
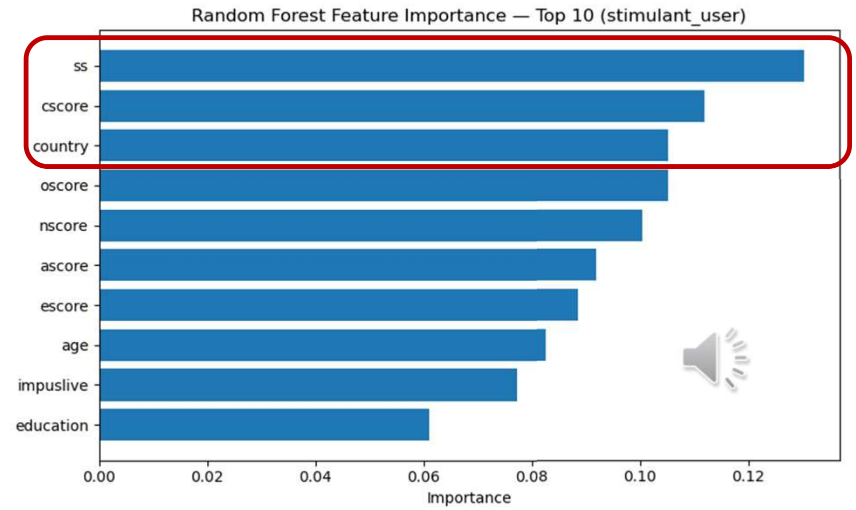
Results: Stimulants

	Logistic Regression	Random Forest
ROC AUC	0.803	0.787
PR AUC	0.745	0.703
Accuracy	72.9%	71.4%
Precision	72.9%	71.4%
Recall	72.9%	71.4%
F1-Score	72.9%	71.4%
Top Features	SS, Country	SS, Oscore, Country





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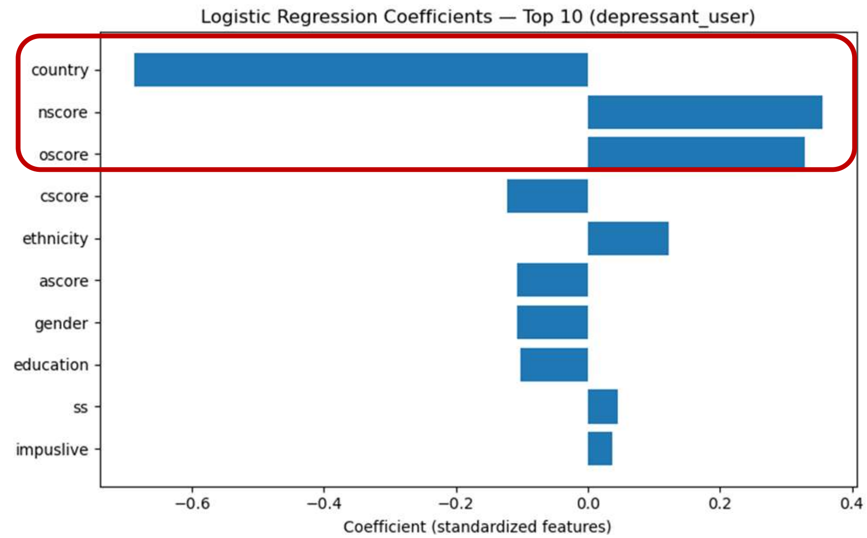
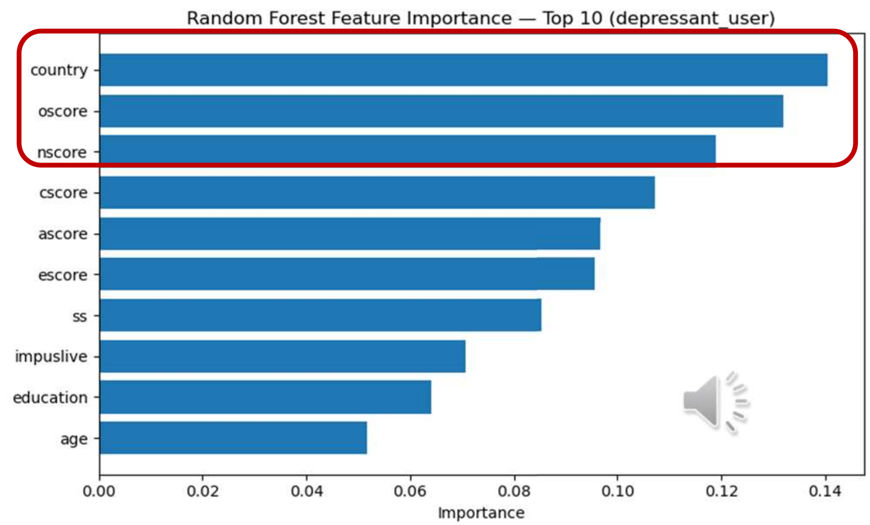
Results: Depressants

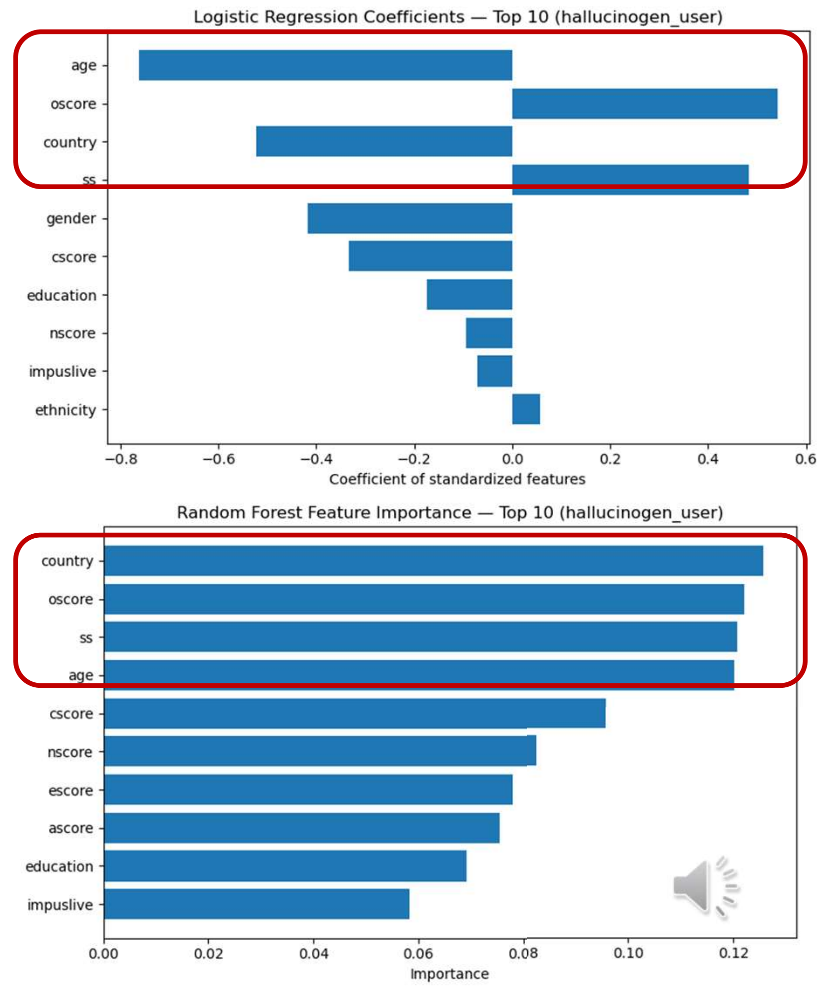
	Logistic Regression	Random Forest
ROC AUC	0.800	0.781
PR AUC	0.737	0.708
Accuracy	72.4%	69.8%
Precision	72.4%	69.8%
Recall	72.4%	69.8%
F1-Score	72.4%	69.8%
Top Features	Country, Nscore, Osocre	Country, Oscore, Nscore





Results: Hallucinogens

	Logistic Regression	Random Forest
ROC AUC	0.861	0.867
PR AUC	0.842	0.851
Accuracy	76.9%	78.8%
Precision	77.4%	78.9%
Recall	76.9%	78.8%
F1-Score	76.8%	78.7%
Top Features	Age, Oscore, Country, SS	Country, Oscore, SS, Age



Policy Implications

- Use personality-based risk screening for prevention targeting high sensation-seeking profiles.
  - Stimulant Users
    - Prevention programs focused on younger, high-impulsivity, high-sensation-seeking individuals
  - Depressant Users
    - Targeted support like mental health counseling or stress management could help for those high in neuroticism.
    - Early screening in high-risk groups might be helpful for early identification of patterns of misuse.
  - Hallucinogen Users
    - Focus on personality traits like thrill-seeking, impulsive individuals
    - Younger demographic

Future Research

INCORPORATE SOCIOECONOMIC, PEER NETWORK, AND ENVIRONMENTAL DATA.

EXPLORE ADVANCED MODELS (E.G., GRADIENT BOOSTING).

VALIDATE RESULTS USING EXTERNAL DATASETS.

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Conclusion

- Personality and demographics can predict drug use.
- Model choice matters by drug category: logistic regression for stimulant and depressant use, random forest for hallucinogen use
- Findings support targeted, data-driven prevention strategies.

Thank you

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References

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